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Part A — User and System Investigation

Determine the username of the account you are currently using.

Display your UID, primary GID, and supplementary groups.

Display all users currently logged in to the system.

Determine what activities the currently logged-in users are performing.

Examine `/etc/passwd` and identify the following information for your current account:

Username

UI

D

GI

D

Home

directory Login

shell

Determine how many user-account entries are present in `/etc/passwd`.

Display only the username and UID fields from /etc/passwd.

Find all accounts whose UID is 0. Explain the security significance of UID 0.

Compare `/etc/passwd` and `/etc/shadow`. Why are password hashes stored in `/etc/shadow` instead of `/etc/passwd`?

```

(prathiksha@kali)-[~]
$ whoami
prathiksha

(prathiksha@kali)-[~]
$ id
uid=1000(prathiksha) gid=1000(prathiksha) groups=1000(prathiksha),999(lpadmin)

(prathiksha@kali)-[~]
$ id -u
1000

(prathiksha@kali)-[~]
$ id -g
1000

(prathiksha@kali)-[~]
$ id -G
1000 4 20 24 25 27 29 30 44 46 100 101 102 114 119 968 979 999

(prathiksha@kali)-[~]
$ who
prathiksha seat0 2026-09-01 21:28 (:0)

(prathiksha@kali)-[~]
$ w
22:02:38 up 34 min, 1 user, load average: 0.30, 0.38, 0.27
USER TTY FROM LOGIN@ IDLE JCPU PCPU
prathiks -ztc/passwd 21:28 0.00s ?

```

```

(prathiksha@kali)~$ grep "^(whoami):" /etc/passwd
prathiksha:x:1000:1000:prathiksha,,,:/home/prathiksha:/usr/bin/zsh

(prathiksha@kali)~$ wc -l /etc/passwd
57 /etc/passwd

(prathiksha@kali)~$ cut -d: -f1,3 /etc/passwd
root:0
daemon:1
bin:2

(prathiksha@kali)~$ awk -F: ' $3 == 0 {print $1, $3}' /etc/passwd
root 0

(prathiksha@kali)~$ cat /etc/passwd
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin

(prathiksha@kali)~$ sudo cat /etc/passwd
[sudo] password for prathiksha:
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin

```

/etc/passwd contains basic user information such as username, UID, GID, home directory and login shell. /etc/shadow stores password hashes and other password-related information and is accessible only to privileged users. Password hashes are stored in /etc/shadow to protect them from being accessed by ordinary users.

Part B — Creating and Verifying Users

Create the following users and ensure that a home directory is created for each:

alice, bob, charlie

Verify that all three accounts have been added to /etc/passwd.

Determine the UID assigned to each newly created user.

Determine the home directory and default shell assigned to alice.

Inspect /home and verify that the required home directories were created.

Set passwords for all three accounts

Switch from your current account to alice using a login shell. Verify the change using an appropriate command, and then return to your original account.

```
(prathiksha@kali)~  
$ ls /home  
alice bob charlie prathiksha  
  
(prathiksha@kali)~  
$ grep -E '^((alice|bob|charlie):)' /etc/passwd  
alice:x:1001:1001::/home/alice:/bin/sh  
bob:x:1002:1002::/home/bob:/bin/sh  
charlie:x:1003:1003::/home/charlie:/bin/sh  
  
(prathiksha@kali)~  
$ id alice  
uid=1001(alice) gid=1001(alice) groups=1001(alice),1004(security)  
  
(prathiksha@kali)~  
$ id bob  
uid=1002(bob) gid=1002(bob) groups=1002(bob),1004(security),1005(developers)  
  
(prathiksha@kali)~  
$ id charlie  
uid=1003(charlie) gid=1003(charlie) groups=1003(charlie),1004(security)  
  
(prathiksha@kali)~  
$ grep -E '^((alice|bob|charlie):)' /etc/passwd  
alice:x:1001:1001::/home/alice:/bin/sh  
bob:x:1002:1002::/home/bob:/bin/sh  
charlie:x:1003:1003::/home/charlie:/bin/sh  
  
(prathiksha@kali)~  
$ grep '^alice:' /etc/passwd  
alice:x:1001:1001::/home/alice:/bin/sh  
  
(prathiksha@kali)~  
$ ls /home  
alice bob charlie prathiksha  
  
(prathiksha@kali)~  
$ sudo passwd alice  
New password:  
Retype new password:  
passwd: password updated successfully  
  
(prathiksha@kali)~  
$ sudo passwd bob  
New password:  
Retype new password:  
passwd: password updated successfully  
  
(prathiksha@kali)~  
$ sudo passwd charlie  
New password:  
Retype new password:  
passwd: password updated successfully  
  
(prathiksha@kali)~  
$ su - alice  
Password:  
$ whoami  
alice  
$ exit
```

Part C — Group Management and Access Control

Create two groups:

security and developers

Add alice and charlie to the security group.

Add bob to the developers group.

Without switching accounts, verify the group membership of all three users.

Display the entries corresponding to security and developers from /etc/group.

Add bob to the security group without removing him from the developers group. Verify that bob now belongs to both groups.

```
(prathiksha@kali)-[~]
$ sudo groupadd security

(prathiksha@kali)-[~]
$ sudo groupadd developers
load average: 0.00, 0.00, 0.00
USER      tty      FROM      LOG      TIME
(prathiksha@kali)-[~]
$ sudo usermod -aG security alice

(prathiksha@kali)-[~]
$ sudo usermod -aG security charlie

(prathiksha@kali)-[~]
$ sudo usermod -aG developers bob

(prathiksha@kali)-[~]
$ groups alice
alice : alice security

(prathiksha@kali)-[~]
$ groups bob
bob : bob security developers

(prathiksha@kali)-[~]
$ groups charlie
charlie : charlie security

(prathiksha@kali)-[~]
$ grep -E '^((security|developers):)' /etc/group
security:x:1004:alice,charlie,bob
developers:x:1005:bob

(prathiksha@kali)-[~]
$ sudo usermod -aG security bob

(prathiksha@kali)-[~]
$ groups bob
bob : bob security developers
```